

Chapter 23

Integrating Energy in Urban Planning in the Philippines and Vietnam

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23.1 Introduction

One of the characteristics of developing countries is increasing urbanization. The Philippines and Vietnam are no exception. In fact, the urban population growth rates of the two countries in 2000–2005 are slightly higher than the average for all developing Asian countries during the same period and significantly higher than the world average (see Table 23.1). Expanding urban population (which to a very large extent is caused by or coincides with industrialization) is one of the factors driving a country's energy demand growth. Conversely, energy is a key force in the process of urbanization and has profound influence on the direction of urban development. In other words, the utilization of energy and the quality of the natural environment in the city cannot be divorced from one other.

All too often, energy issues are not taken into account or given priority in the planning of cities. Indeed, energy services are not singled out as among the critical issues in urban environmental governance (see, for example, UNESCAP (2005)) even if energy production, transformation, and use are recognized as the main contributors to air pollution and greenhouse gas (GHG) emissions. The traditional concept used in the formulation of the development plans of cities is that urban space is planned and designed to be livable. The human is said to be the focus of all city development planning: the overall concern is the satisfaction of the needs of the city population and the promotion of human dignity through constant improvement of people's welfare and health. Although adequate and reliable energy supply and clean and healthy environment contribute to man's well-being, energy issues are not fully addressed in this traditional concept of urban planning. This is ironic, given the severely deleterious impact on both health and well-being conventional urban energy systems can have on a city's population.

There are several reasons why many cities have not integrated energy in urban development planning. Among these are:¹

¹This is adapted from a similar project proposal written by Mr. Manuel Soriano, Regional Technical Advisor-Climate Change, UNDP/GEF.

Table 23.1. Urbanization trends in selected developing Asian countries.

Countries	Mid-year population (million)		Population growth rate (%)	Urban population as % of total population		Urban population growth rate (%)
	2000	2005		2000	2005	
China	1267.4	1307.6	0.6	36.2	43.0	4.1
Cambodia	12.6	13.8	1.9	16.0	17.7	4.5
Indonesia	205.8	219.9	1.3	42.0	48.1	4.1
Lao PDR	5.2	5.6	1.4	19.3	21.6	3.8
Malaysia	23.5	26.1	2.2	62.0	63.0	2.5
<i>Philippines</i>	76.9	85.2	2.1	58.5	62.7	3.5
Thailand	62.2	64.8	0.8	19.0	32.5	12.2
<i>Vietnam</i>	77.6	83.1	1.4	24.2	27.0	3.6
Bangladesh	128.1	137.0	1.4	23.2	25.1	3.0
India	1019.0	1107.0	1.7	27.7	28.7	2.4
Nepal	22.6	25.3	2.3	13.4	15.8	5.7
Pakistan	139.8	154.0	2.0	33.0	34.0	2.6
Total	3289.0	3495.0	1.2	33.9	37.7	3.4
Developing Asia						
World	6085.6	6464.8	1.2	46.7	48.7	2.1

Source: ADB 2006.

- Energy suppliers control the energy market in cities. This is because energy supply in these cities is dependent on the market planning of the energy suppliers, which, except for electric utilities, are usually a private commercial concern.
- Only the energy (mainly electricity) requirements of the development programs and local government operational activities are covered in the urban development plans of many cities. This is because there is no perceived need for energy planning in cities, thus energy planning has not been a local government concern. Energy planning is a national government concern in many if not all developing countries.
- City planners and officials are not aware of the direct links between energy and urban development.
- Environmental concerns are addressed separately by cities. This is because energy–environment links to spatial and development planning is not well known among city planners and local officials.
- There is no information about the magnitude of energy usage and consumption trends in many cities as energy supply and consumption surveys are not conducted in these cities.
- There are neither regulations nor official policies on the practice of energy conservation in many cities. This problem is due to the generally low level of awareness about energy conservation among city officials, city planners, and the general public. The low level of awareness is partly due to the general lack of awareness campaigns on energy conservation in many cities.

Naga City in the Philippines and Can Tho City in Vietnam were ‘host cities’ to an EU-funded project that had aimed at preparing the integration of energy concerns in urban

planning in ASEAN cities.² Naga City and Can Tho City represent a small city and large city respectively, operating under different political systems of government with balanced yet diverse economic structures. Both exhibit high economic growth yet different levels of economic development, and a large potential for integration of energy in urban development planning – albeit at different stages in their evolution to more sustainable forms of local governance. The present article chronicles the experience of the two cities in that project. Specifically, it describes a methodology for integrating energy in urban planning, highlights key conditions for its successful application, and presents some results.

23.2 Socioeconomic Background of Naga City and Can Tho City

23.2.1 Naga City

Naga City is located within the heartland of the province of Camarines Sur. It is 377 kilometres (km) south of Manila and 100 km north of Legazpi City, the capital of the Bicol region (or Region V). Occupying a total land area of only 84.48 square kilometres (sq. km) and with a population of only around 150 000, Naga City may be considered a small city by national and regional (Asian) standards. The city comprises 27 *barangays*³ (or villages) and remains predominantly agricultural (accounting for 75% of total land use). A big portion of the population (more than 60%) is within the productive 15–64 years old bracket. The population of the city of Naga grows at a rate of 1.65% per year and is expected to double in 42 years. Like many cities in the Philippines, and probably in many parts of the globe, Naga City's population expands during school months and national holidays, as well as during working hours of the day.⁴

Naga City is the commercial, educational, as well as religious centre of the Bicol region. It is one of the fastest growing economies of the country with 6.5% annual growth rate. Naga captures 21% of the whole investments in the Bicol region and was once chosen as the Most Business Friendly City by the Philippines Chamber of Commerce and Industry. The city's commercial and industrial activities are largely related to agricultural production. Major crops produced by the agricultural sector include rice, sugarcane, coconuts, fruit trees, and corn, but their values of production have been on the decline. Backyard livestock and poultry production shares the biggest portion of total value of agricultural output. The average family income of Naga City's residents is 126% higher than the average in Bicol and 42% higher than the national average.

23.2.2 Can Tho City

Can Tho City is located at the centre of the Mekong river delta and 160 km south of Ho Chi Minh City, Vietnam's biggest city. Can Tho City, on the other hand, is the biggest city in

²The project was funded through the EU-ASEAN Energy Facility, which was executed from Jakarta, Indonesia by the ASEAN Centre for Energy (ACE). From 12 initially surveyed cities, the project selected three ASEAN cities – Naga City, Can Tho City, and Luang Prabang in Lao PDR – mainly because of their strong willingness to cooperate in the project and they could be good examples under different economic, social and political conditions. However, due to time and space constraints primarily, Luang Prabang is not included in this article. Second, the case of Naga City and Can Tho City may be viewed as examples from a small city and large city, respectively, with balanced economic structure. Luang Prabang is a medium-sized city that is heavily dependent on tourism.

³The *barangay* is the smallest administrative unit of government in the Philippines.

⁴Many residents from neighbouring municipalities or towns work or study in Naga City during the day.

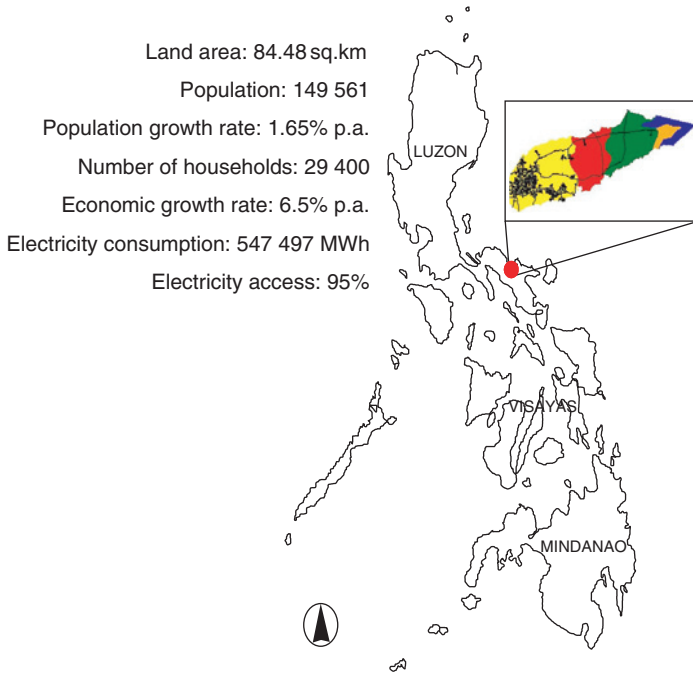


Fig. 23.1. Location of and basic facts on Naga City, Philippines.

the Mekong delta region, occupying a total land area of 1390 sq. km and with a population of 1.127 million (see Fig. 23.2). The city is divided into four urban districts and four suburban districts. But more than 80% of Can Tho City's land area remains dedicated to agriculture. Like Naga, a big portion (69%) of the total population is within the 15–64 years old bracket.

But more than Naga City, Can Tho City's economy is growing at the rate of 14.95% per year. Most of this growth comes from commercial or trade activities, which in terms of number of establishments, are growing at 33.1% per year and account for 43.7% of local economic output. The number of industrial establishments is also growing significantly at 5.3% per year, and industrial output contributes 35.1% to local economy. But agriculture remains a very important sector. It accounts for the remaining 21.2% of local economic output and employs 37% of the labour force, even more than those employed in the industrial and commercial (trade) sectors combined.

23.3 Background on the Two Cities' Urban Planning Process

As expected, both Naga City and Can Tho City have dedicated departments responsible for urban planning and development. In Naga City, like in all other cities and municipalities in the Philippines, urban planning falls within the responsibility of the City Planning and Development Office (CPDO). In Can Tho City, urban planning and development is the responsibility of the Department of Planning and Investment (DPI), which is an extension of the country's Ministry of Planning and Investment at city level. Except for nature conservation, which was reportedly not covered by Can Tho City, both Naga City's CPDO

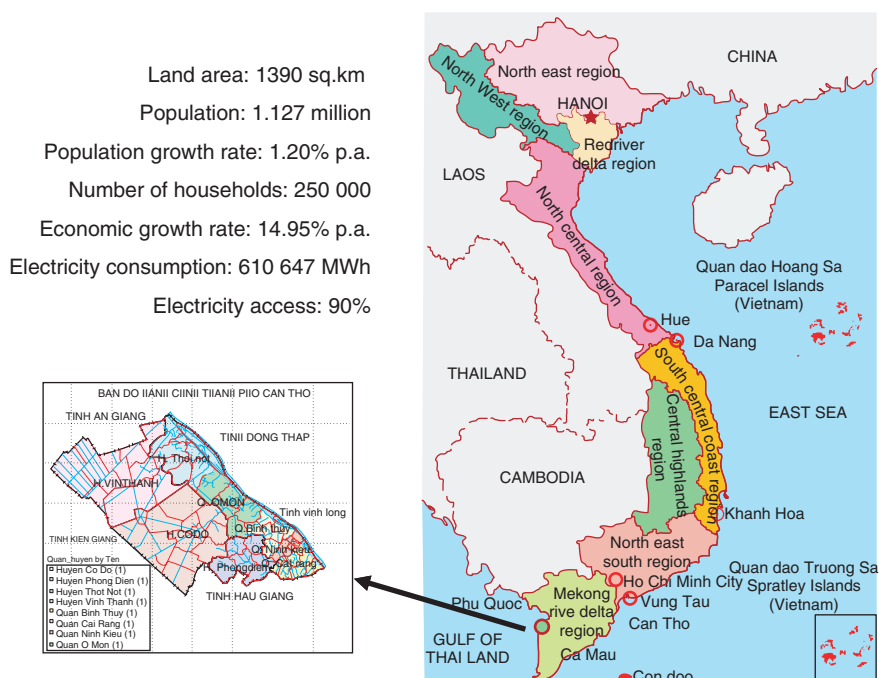


Fig. 23.2. Location of and basic facts on Can Tho City, Vietnam.

and Can Tho City's DPI cover a wide range of urban planning sectors, including housing, water supply, roads, transport and traffic, waste management, land management, energy supply, energy demand management, and industrial zones. It is important to note that both cities claim to be dealing with energy supply and energy demand management in their urban planning.

However, urban planning is a multi-sectoral and multi-stakeholder process in Naga City, whereas Can Tho City reports that other agencies are not involved in the city's urban planning. Figure 23.3 shows the steps in urban planning in Naga City. Table 23.2 lists the agencies and other organizations representing each of the sectors covered by urban planning in Naga City. In fact, Naga City has received more than 140 citations from national and international bodies for its success in innovative and participatory governance. Box 23.1 summarizes the Naga City 'model'.

Nevertheless, while this model has given Naga City considerable success in urban planning, the two cities admit a number of weaknesses that tend to be possessed by cities in developing countries:⁵

- Data gaps – lack of accurate and up-to-date data banking.
- Lack of adequate technology (e.g. computers, training).
- Lack of communication/coordination among planning bodies.

⁵The project finds more or less these same weaknesses in all the 12 cities surveyed at the start of the project.

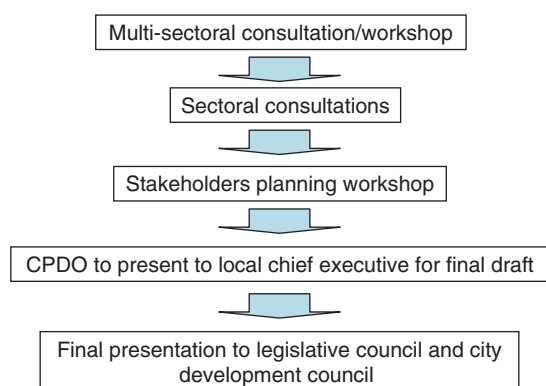


Fig. 23.3. Urban planning in Naga City.

Table 23.2. Other sectors and stakeholders involved in Naga City urban planning.

Other organization involved	Responsibilities and sector
● Housing and Land Use Regulatory Board (HLURB) ● Housing and Urban Development Coordinator Council (HUDCC) ● Urban Poor Affairs Office (UPAO) ● Housing and Urban Development Board (HUBD) ● Department of Public Works and Highways (DPWH) ● City Engineer's Office (CEO) ● Department of Environment and Natural Resources Office (DENR) ● City Environment and Natural Resources Office (CENRO) ● Solid Waste Management Board (SWMB) ● LANCO (MRF) ● Protected Area Mgt Board (PAMB) ● DENR ● CENRO ● Metro Naga Chamber of Commerce and Industry (MNCCI) ● Naga City Investment Board (NCIB) ● Investment Promotions and Action Center (IPAC) ● Metro Naga Water District (MNWD) ● Phil. National Police (PNP) ● Transport and Traffic Management Office (T&TMO) ● Department of Agrarian Reform (DAR) ● HLURB ● Camarines Sur Electric Cooperative II (CASURECO II) ● Camarines Sur Electric Cooperative II (CASURECO II)	● Housing ● Roads ● Waste management ● Nature conservation ● Industrial zones ● Water supply ● Transport and traffic ● Land management ● Energy supply ● Energy demand and management

- Duplication of task/functions.
- Lack of monitoring of different projects and programs.
- Level of technical skills/knowledge of personnel involved in urban planning are not adequate.

Box 23.1.
The Naga City ‘model’

Three elements form the foundation of Naga City governance model: (1) Progressive Development Perspective; (2) Functional Partnerships; and (3) People Participation.

The first element is the philosophy that anchors all development efforts and seeks to mainstream all sectors of society in accepting their role in local development. This is a function of leadership that the local administration must provide. This element is also associated with the concept of ‘growth with equity’ that has been embraced by Naga City as its core philosophy. ‘Growth with equity’ seeks to promote economic development and to sustain the implementation of pro-poor projects to ultimately build prosperity for the community at large.

The second element encompasses the vehicles that enable the city to tap community resources for priority undertakings, in the process multiplying its capacity and enabling it to overcome resource constraints that usually hamper government. The city’s partnership mechanisms can be between the local government unit (LGU) and other community groups or government agencies, or between the LGU and individuals. They can be for growth programs or equity-building strategies. They may also be government initiated or private sector/community led.

The last element includes mechanisms that ensure inclusion of individuals and the community in government decision-making. They promote long-term sustainability by generating broad-based stakeholdership and community ownership over local undertakings. One such mechanism is the landmark local legislation ‘Empowerment Ordinance of the City of Naga’ that established the Naga City People’s Council (NCPC). NCPC is composed of all accredited business, non-government, and people’s organizations within Naga City. NCPC has representatives in all special bodies in the city and participates in all decision-making activities of the city. The latest mechanism to enhance citizen’s involvement in governance is the *i*-Governance program of the city. *i* stands for information openness, interactive engagement, inclusive governance, and innovative management. *i*-Governance focuses on the individual member of the community. It has two basic tools: the Naga City website and the Naga Citizens Charter.

Source: Extracted from (Robredo and Jesse 2003).

23.4 Energy-Integrated Urban Planning (EIUP): Introduction to the EIUP Methodology⁶

One of the objectives of the project was to introduce a methodology for integrating energy strategies in the selected cities with high potential for replication in other ASEAN cities. In this regard, the project used a methodology that was labelled EIUP (energy-integrated urban planning). EIUP is a series of activities or steps that will be undertaken by the cities to help them define their short- and long-term strategies and action plans in solving

⁶The methodology was developed by Dr Gerhard Weihs of Centric Austria International (CAI), the lead partner for the EU-funded project involving the two cities. This section derives from the Final Technical Report of the project.

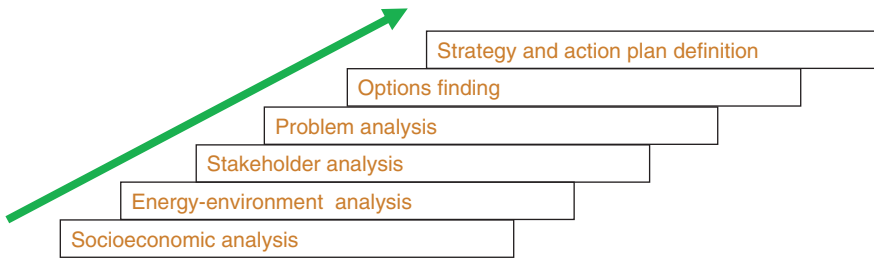


Fig. 23.4. The EIUP methodology.

energy problems that can be addressed at the local level. It comprises the activities as shown in Fig. 23.4.

The EIUP is guided by the following principles:

- EIUP should be based on a reliable database.
- EIUP should be in line with other policy and legal frameworks both at city and national levels.
- EIUP should be interrelated with other planning sectors.
- EIUP should involve all stakeholders and generate benefits for the whole citizenship.
- EIUP should contribute to sustainable development.
- EIUP should respect the local conditions (demands, constraints, resources).
- EIUP should identify feasible and justified improvements.
- EIUP should prepare the reliable ground to take concrete action.

The EIUP uses a concrete tool, the EIUP Working Report, which is a visible guide for undertaking the steps and activities of energy-integrated urban planning, monitoring the progress of each step, and documenting the results. The EIUP Working Report is structured into seven chapters that parallel the steps and activities in EIUP:

1. Socio-economic city profile
2. Energy and environment
3. Stakeholder analysis
4. Definition of problems and targets
5. Option finding
6. Action plan
7. Summary

Chapters 1 and 2 are meant to provide sufficient information to describe the existing situation in social, economic, environmental and energy-related aspects, from which the city starts its efforts of EIUP. Chapters 3 and 4 have an analytic character and identify the scope of required integration in the context of pressing problems. Chapters 5 and 6 identify and clarify the options in terms of their feasibility and implementation aspects. Chapter 6, in particular, provides clear recommendations to take action according to clear priorities and feasibility. Chapter 7 presents a summary that can be used for communicating the results of EIUP to decision-makers, stakeholders, and third parties.

The EIUP Working Report is a very comprehensive document and requires very comprehensive information, especially if the recommended actions mean justifying investment

or raising capital. It is likely that not all information required by the EIUP Working Report is readily available in the cities. Data and information can be added step by step in the future, as they become available. Indeed, some data required by the EIUP Working Report may not be available at the city level, but can be obtained from national agencies or relevant institutions. For example, the data required to build the energy balance table may be available from various sources and institutions. In this case, it may be necessary to devise a way of collecting, organizing, and synthesizing those data. But it is important to get the necessary data before implementing any of the proposed actions.

Lastly, but very important, the EIUP Working Report should be prepared by a team, not by a single person, in the same way that EIUP is coordinated by a team of local technical experts. This team, acting as a technical working group, should work on data collection first; then answer main questions, discuss and rank individual city problems, identify priorities and targets, explore options, and draw strategy and a feasible action plan.

23.5 EIUP: Motivations and Conditions for Success

23.5.1 Host city motivations

Naga City expected EIUP to fulfil its promise – to provide good inputs to city development planning. At the same time, Naga City expected that EIUP would enhance local governance. With an international reputation of effective local governance, Naga City wanted nothing less from EIUP.

In the case of Can Tho, the EIUP was seen as providing opportunities for developing indigenous renewable energy sources. Indeed, Can Tho has huge potential for developing renewable energy resources because of its large agricultural base and favourable location that lends itself to solar energy development.

Both Naga City and Can Tho were conscious that the growing energy demand in their respective cities could cause deterioration in the local environment. The two cities recognized, for example, that their mounting solid waste problem is caused by increasing material and energy consumption. They understood that inefficient energy supply and use leads to deterioration of energy services and further worsens environmental conditions. They saw that the increasing energy demand in the transport sector is causing increase in local air pollution as well as carbon dioxide emissions more than any other sector. The two cities expected EIUP to provide solutions to their environmental problems.

23.5.2 Political commitment (to environment-friendly development)

This project would not have been made possible without the strong commitment from the highest political leadership of the two cities. The project sought to formalize this commitment through an agreement, a Memorandum of Understanding (MOU), between the host city and the international partners. The MOU, on the one hand, committed the international partners to conduct the planned activities of the project and, on the other hand, committed the local partners or host cities to actively participate in those activities. Through the MOU, the city and the international partners agree to successfully deliver the promised benefits of the project together.

To be sure, this strong political commitment emanates from the presence of effective and visible environment-friendly local governance in both cities. Naga City even declares this environment-friendly governance in its vision statement (see Box 23.2). But more importantly, the two cities practise it by dedicating an office that focuses on local environmental

Box 23.2.
Naga City vision statement

‘An Maogmang Lugar’⁷

We envision Naga

As home of God-loving, happy people

- Who are child-friendly, gender fair and globally competitive
- Rooted in positive moral and cultural values, and
- Belonging to strong, unified and mainstreamed sectors

As a progressive and peaceful community

- Considered a centre of people-focused education, development and governance
- With a clean, healthy and sustainable environment, and
- An economy that provides livelihood opportunities for all, ensuring equitable distribution of wealth and proper utilization of resources.

Source: Naga City Citizens Charter, second edition, August 2004, p. 15.

management, through the regular delivery of environmental services and actual execution of local environmental initiatives, and by continuously building their capacity in environmental management through active participation in international urban environment networks.

23.5.2.1 *Dedicated office in urban environmental management*

Within Naga City Hall is the 68-man strong Environment and Natural Resources Office (ENRO), which claims to be the largest department in the City Hall. On a regular basis, ENRO offers and has ‘pledged’⁸ to provide the following frontline services (City of Naga 2004):

- Receive, record and file communication and applications for environmental certificates.
- Prepare quarrying permits, permits to cut trees, and other environmental certificates.
- Sign and issue permits and certificates upon approval of the local chief executive (LCE), or the mayor.
- Attend to requests related to solid waste management, including: (1) responding to garbage collection complaints; (2) garbage collection schedule; and (3) request collection of garbage.
- Attend to requests related to watershed management, including: (1) request for grass cutting and trimming of trees; (2) assistance for tree planting activities; (3) inspect/monitor quarrying activities; (4) proposed or actual construction of building.

⁷This literally means ‘A Happy Place’.

⁸Each major department or office of the Naga City Hall has declared ‘Performance Pledges’, which is a public statement, embedded in the Naga City Citizens Charter, to perform their functions with ‘the highest possible service performance’.

- Attend to requests related to Socio-Cultural Management Programs, including: (1) information education campaign for ENRO; (2) assist in the preparation of training modules and conduct of training desired; (3) link clients to entities and agencies concerned.
- Respond to garbage collection complaints.
- Clean city street gutters and pick-up points and maintain cleanliness daily.
- Collect garbage in pick-up points based on schedule and area coverage.

It should be noted that each local government unit in the Philippines has been mandated to be responsible for environmental management at their level since 1991 through the Local Government Code. Thus, it is expected that each would set up a similar office as Naga City's ENRO. However, garbage collections, by far the most important environment management-related function of a city or municipality, remain usually with the Engineering Office of many municipalities and cities, and one or few staff under an existing department (for example, the city planning and development office, which is present in all LGUs) maybe given the task of all other environment management-related functions.

In Can Tho, urban environmental management is handled by the Department of Science and Technology of Can Tho (DOST). Below is a list of the present responsibilities and undertakings of DOST:

- Evaluating and reporting to the Chairman of the People's Committee on concrete regulations, policies, and strategies regarding science, technology and the environment, in harmony with the local characteristics and not contravening the regulations and guidelines of the central and provincial government.
- Setting up and submitting to the Chairman of the People's Committee the decisions on tasks regarding science, technology, and environmental protection development programs in accordance with the social and economic development demands of the province.
- Providing guidelines for science organizations to formulate and develop plans to apply new technology to production units and residential developments; summarizing and submitting to the People's Committee for decision, supervision and the implementation of development programs and projects funded by the government budget.
- Managing the technology transfer activities; taking part in the government's appraisal of new technology for important investment projects in the province. Supervising and guiding the application of new technology into the development projects of the provincial economic production organizations.
- Formulating plans and local environmental protection methods in accordance with laws and regulations. Following upon and supervising the implementation of environmental protection measures. Presiding, appraising, and commenting on environmental impact assessment reports, monitoring investment projects belonging to the provincial authority; checking and supervising the environmental activities in the province.
- Monitoring Standard – Methodology – Quality for manufactured products and commodities in accordance with laws and regulations.
- Managing the activities regarding industrial ownership according to the industrial ownership protection decrees and other legal documents.
- Implementing the control function in executing regulations, guidelines and laws dealing with science, technology and the environment. When the inspection is required, submitting the results to the People's Committee for approval.

23.5.2.2. *Local initiatives and accomplishments*

Both ENRO of Naga City and DOST of Can Tho have led the actual implementation of a number of local environmental initiatives, beyond their regular functions, to further urban environmental management.

Naga City owns and operates the Balatas Material Recovery Facility (MRF) which processes 40–50 tons of garbage daily⁹ and produces organic fertilizers in the form of soil enhancers. Through the sale of the fertilizer, the MRF earns for Naga USD168 000 per year (International Council for Local Environmental Initiatives (ICLEI) 2006), as well as secures the employment and livelihood of close to 155 families (Robredo and Jesse 2006). The City's Clean Air Initiative has succeeded in continuously improving local air quality. For example, the city reported that average TSP readings had further improved to 80.6 µg/NCM in 2006 (Robredo and Jesse 2007), from 110.65 µg/NCM in 2003, sustaining its good air quality rating. The city's water quality improvement program has improved water quality of the Naga River, maintaining its Class C grade, which means that it could be used for irrigation and navigation purposes. Naga City through the Isarog Garden Society, a local NGO, also maintains an Eco-Park.

Can Tho City has been chosen to become a model of Sustainable Urban Management by the Prime Minister of the People's Republic of Vietnam, who takes a strong and personal interest in the promotion of environmental issues. (In fact, he has directed the individual provincial governments to give priority to protection of the environment.) With assistance from Green City Denmark, Can Tho City has organized the Green Ambassadors that increase the public's awareness and encourage participation in the city's solid waste management program.

23.5.2.3. *Active participation in international urban environment networks*

The two cities' political commitment in environment-friendly local governance is also demonstrated by their active participation and membership in international urban environment networks that aim, among other goals, to increase the capacity of cities in urban environmental management.

Naga City is a prominent member of the International Council on Local Environment Initiatives (ICLEI) and a pilot city of ICLEI's Cities for Climate Protection Campaign (see Box 23.3). ICLEI provides the city technical support in the formulation, adoption, and monitoring of measures that would reduce GHG emissions. Through ICLEI, Naga also establishes links with other cities and organizations.

Can Tho City, on the other hand, is one of the few Asian member-countries of the Green City Network of Green City Denmark (GCD). Through its membership in this network, Can Tho City has received technical and financial assistance from GCD in undertaking solid waste management.

23.5.3 *Organization of local EIUP teams*

EIUP is designed to be carried out by a team, not by a single person. Another key therefore to the successful implementation of EIUP is the formation of a team of dedicated and competent local experts that will carry out the activities of the EIUP, including collecting and analysing data, liaising or coordinating with key stakeholders, identifying problems and

⁹ But the facility can handle 60–70 tons of municipal solid wastes daily.

Box 23.3. About ICLEI

ICLEI was founded in 1990 as the International Council for Local Environmental Initiatives. The council was established when more than 200 local governments from 43 countries convened at its inaugural conference, the World Congress of Local Governments for a Sustainable Future, at the United Nations in New York. Now, ICLEI–Local Governments for Sustainability is an international association of local governments and national and regional local government organizations that have made a commitment to sustainable development. More than 475 cities, towns, counties, and their associations worldwide comprise ICLEI’s growing membership. ICLEI works with these and hundreds of other local governments through international performance-based, results-oriented campaigns and programs. One such program is the Cities for Climate Protection Campaign. Established in 1993, ICLEI’s CCP works with local governments to improve urban management and address economic, environmental and social concerns. In doing so, local governments achieve the side benefits of reducing air pollution, waste, and greenhouse gas emissions. In 1999, the Cities for Climate Protection Campaign began with five pilot cities in the Philippines. In 2001–2003, funded by USAID and CIDA, CCP activities expanded from the Philippines to cover Indonesia and Thailand. Indonesia started with three cities in 2001 and has grown to ten in 2004. Thailand has six cities participating in the Campaign. The number of CCP participants in the Philippines has grown to 14 in 2005. Naga City is among ICLEI’s members in the Philippines.

(See also <http://www.iclei.org/index.php?id=594>)

Source: ICLEI website, <http://www.iclei.org>



Fig. 23.5. Green City Network.

Table 23.3. Composition of the local EIUP team in Naga City.

Institution and position within this institution	Major concerns and tasks within the EIUP team
City councillor, team leader	Legislate ordinance
Chief of City Environment Regulation Office (ENRO), asst. team leader	Executive implementation
Socio-Cultural Division Chief, City ENRO	Assist in planning and implementation
Planning officer IV, City Planning and Development Office (CPDO)	Assist in planning
Project officer I, CPDO	Planning
City engineer, City Engineer's Office (CEO)	Regulate project implementation
Engineer II, CEO	Assist in project designing
Illustrator I, City ENRO	Design projects
Program director, Naga City People's Council (NCPC)	Support planning development
NGO staff, NCPC	
Technical writer, City ENRO	Technical writing

priorities, and recommending options. This technical working group is headed by a leader that will report the progress and results of the EIUP to the local chief executive – Mayor of Naga City or Chairman of Can Tho People's Committee. In Naga, one of the city councillor's was appointed to lead the local EIUP team, but he was assisted by the Chief of ENRO, which is the unit responsible for the city's environmental management. This arrangement actually proceeded from the established practice of the city to share the executive functions with members of the legislative council, particularly on ad hoc committees. But the technical work of the Naga City EIUP team was still led by the Chief of ENRO, while the city councillor actually performed an advisory role. Representatives from the City Planning and Development Office, City Engineer's Office, and Naga City People's Council, which is an umbrella organization of all the active local NGOs in Naga City, completed the Naga City local EIUP team.

Similarly, the Vice-Chairman of the Can Tho People's Committee, which performs both executive and legislative functions, was part of the local EIUP team as an advisor, and the Chief of the DOST, which is responsible for the city's urban environmental management, was the designated team leader of the technical working group. Representatives from the Department of Planning and Investment, Department of Industry (which oversees the energy sector), Department of Natural Resources and Environment, and the Can Tho City People's Committee completed the local EIUP team for Can Tho. Representatives of other agencies and the energy companies were part of the local EIUP team on an ad hoc basis.

While it seems that the Can Tho EIUP team had more diverse representation from government agencies than that of Naga City, the formation of the local EIUP teams was dependent on the organizational structure of each city as shown earlier. What is important in the organization of the local EIUP team, as demonstrated in the experience of Naga City and Can Tho, is that key stakeholders are adequately represented in the team. This is essential for accomplishing the purpose of EIUP. In the Naga City EIUP team, the local legislators, the economic planning sectors, the infrastructure sector, and the NGOs were represented. In the Can Tho EIUP team, the economic planning sectors, industry and energy, environment, and infrastructure sectors were represented.

Table 23.4. Composition of the local EIUP team in Can Tho.

Institution and position within this institution	Major concerns and tasks within the EIUP team
Vice-Chairman	Project supervisory role
People's Committee of Can Tho	
Director, Department of Science and Technology	Team leader
Vice Director, Department of Planning and Investment	Support information about related planning
	Considering and ratifying projects
Vice Director, Department of Natural Resources and Environment	Technical support
Chief of Technology Bureau, Department of Industry	Technical support
The main Expert of People's Committee Office	Administration
Chief of Environment Monitoring Station, Department of Natural Resources and Environment	Technical support
Director, Can Tho Power Company	Ad hoc basis
Director, Tay Nam Bo Petrol Company	Ad hoc basis
Manager of Technical Division, Department of Transportation and Public Works	Ad hoc basis

23.6 Application of EIUP: Key Results for Naga and Can Tho

23.6.1 Opportunities in energy and environment

Through the application of EIUP, Naga City saw the following opportunities:

- Potential for development of decentralized RES systems.
- National government supports power supply companies to deal with service improvement.
- Political support from the city government to energy efficiency and RES.
- National legislation supports development of local energy resources through various fiscal and non-fiscal incentives.
- Possibility to benefit from local energy resources – non-power and power related in the case of electricity.
- Improvement of quality of energy supply.
- High energy prices will promote fuel switching and implementation of new technologies.
- Requirement of national government to use biodiesel to replace a part of oil fuels in transport (1%) in the case of government agencies.
- Waste-to-energy project development and implementation.

Can Tho, on the other hand, became more convinced that urbanization guided by the principles of sustainable development will:

- Promote environmental protection and improvement in the government and communities;
- Improve transportation systems, architectural landscape, and manufacturing technologies; and

- Ensure support from the central government for improving the city's energy production and use.

23.6.2 Stakeholder analysis

Naga City and Can Tho had the common opinion that practically all the broad sectors of their local economies are the stakeholders of EIUP (see Table 23.5). Unsurprisingly, the common expectation or interest of these sectors was for EIUP to achieve an adequate, reliable, and low cost or reasonably priced and affordable energy supply, with minimal impacts on the environment. The stakeholder analysis also confirmed the potential of some sectors for developing and investing in renewable energy sources and energy efficiency. Equally importantly, the stakeholder analysis stressed the role of other sectors (particularly the general public and final consumers) in policy development, implementation and monitoring, as well as in increasing the social acceptability of energy projects (if the stakeholders are involved in the early stages of project development). At the same time, the stakeholder analysis identifies the lack of awareness and capacity on technologies, finance, and regulation as permeating most, if not all, sectors. These deficiencies could be addressed through the EIUP.

The stakeholder analysis included the identification of the responsible bodies for the different planning sectors and the related laws and policies and their target sectors in each of the two cities. These provided the institutional, policy, and legal context of EIUP that could support or limit the feasibility of its results or recommendations. For example, Naga City should comply with the Philippine Energy Plan that sets direction and targets related to energy supply and demand, and the Electric Power Industry Reform Act that outlines the new policies in the electricity sector. On the other hand, national environmental laws provide opportunities for Naga City in exploiting waste-to-energy potentials. In the case of Can Tho, however, the lack of clear directions on renewable energy development and energy efficiency and conservation at the state level tends to limit the identification and assessment of clean energy options at the level of local government.

23.6.3 Problem analysis

The SWOT analysis was necessary for identifying the real energy concerns of the two cities. The stakeholder analysis was necessary for identifying the sectors affected or affecting the problems and eventually engaging them to jointly rank and prioritize these problems and analyse their impacts through a consultation. Both cities went through the latter process without direct intervention from the international team of experts, except undergoing a small group workshop with selected members of the local EIUP team to become familiar with the steps in problem analysis.¹⁰

The stakeholder consultations in the two cities resulted in the identification of priority problems and their ranking as shown in Table 23.6.

Except for the availability of a reliable energy (electricity) supply and to some extent the use of renewable energy sources, notice the contrasting priorities of Naga City and Can Tho when it comes to energy issues. In particular, while Can Tho gave greater importance to having an energy policy than efficient energy end-use, Naga City gave more importance

¹⁰This workshop was conducted during the study tour in Europe when representatives from the three host cities and the EU and ASEAN partners exchanged experiences with stakeholders from selected EU cities and organizations.

Table 23.5. Stakeholder analysis of Naga City and Can Tho.

Stakeholders	Interests and expectations on EIUP	Potential	Deficiencies
Agriculture	Electricity supply at low cost	Large biomass resource	Lack of awareness on technologies and financial resources
Residential	Reliable and affordable energy supply; reduced environmental impacts of energy production and use		Lack of awareness on energy conservation and renewable energy resources
Commercial, trade, and industrial sectors	Adequate, reliable, and reasonable cost energy supply; opportunities for producing or selling electricity	Participation in EIUP development and monitoring; application of clean energy technologies; ability to invest in energy production and energy efficiency and savings	Lack of awareness on technologies, regulatory framework, monitoring and enforcement
Utilities and energy suppliers	Development of renewable energy resources; efficiency in electricity distribution and end-use; distributed generation	Support for energy efficiency programs; cooperation in electricity business; rehabilitation and extension of distribution network; big (and established) market on fossil fuel consumption	Few or inadequate expertise in developing renewable energy resources and energy efficiency programs; lack of information on fossil fuel end-use consumption
NGOs/POs	Reliable, accessible, and low priced energy supply; reduced environmental impact of energy use	Involvement in policy development, implementation, and monitoring; preparation of regulations; involvement in social preparation of projects	
Local government	Develop the tool for future improvement of energy supply, reduction of energy costs, improvement of the environment, improvement of standard of living, create climate for business development and employment	Develop capacity for energy planning; conduct awareness campaign for general public; facilitator of implementation of activities identified in the Action Plan; provider of co-financing of projects	
Other government agencies (public sector)	Monitor implementation of other or related government policies		Possible lack of coordination between the city government and government offices; lack of support to implementation

Table 23.6. Problem storage (ABC Analysis) for Naga City and Can Tho.

Priority problems	Ranking
Can Tho	
1. Lack of energy development planning and policy	A
2. Shortage of energy	A
3. Inefficient energy end-use	B
4. Lack of know-how on sustainable waste-to-energy and other renewable energy technologies	B
Naga City	
1. Reliability of electricity supply	A
2. Lack of comprehensive policies and energy management	B
3. Insufficient use of alternative energy sources	B
4. Not all communities are connected to power supply	C
5. Insufficient use of alternative motor fuels	B
6. Insufficient extension of energy conservation and audits	A

Note on ranking:

A = problem with big pressure and urgent need for solution.

B = problem of great importance, which should be solved in the medium term.

C = important, but does not need urgent solution.

to extending its energy conservation efforts rather than addressing the lack of comprehensive energy policy. These differences in energy priorities between the two cities could be explained by the differences in the stage of their economic development. Although Can Tho is 16 times larger and seven times more populous than Naga City, the poorest segment of this Philippine city is more than five times richer than the equivalent portion of Can Tho City's residents. This also explains why the per capita electricity consumption of Naga City is higher than Can Tho, thus actually providing more opportunities for energy efficiency and conservation. Moreover, with a population of more than one million and lower electrification rate, there are actually more people in Can Tho without access to electricity than in Naga City, justifying the higher importance given by Can Tho to availability of energy supply compared to Naga City's concern on supply reliability. (It is estimated that about 112,000 Can Tho residents do not have access to electricity compared to Naga City's 7,500 residents.)

In options finding, these problems are translated into concrete objectives. These objectives are then explored as to possible activities for achieving them, the obstacles to the realization of these activities, recommendations for overcoming the barriers, and concrete first steps.

23.6.4 Strategy and action plan

Options finding is also a bottom-up approach in defining the cities' EIUP strategy and formulating the final plan of actions in solving the problems identified earlier. The EIUP strategy is a statement of the city's energy policy, vision, mission, and goals. These are shown in Box 23.4 and Box 23.5 for Naga City and Can Tho, respectively.

The actions identified to address the priority problems mentioned and described earlier are also listed in Boxes 23.6 and 23.7.

Box 23.4.
EIUP strategy of Naga City

EIUP Policy:

To overcome identified barriers, the focus of the EIUP will be:

- Implementation of energy planning activities, establishing suitable policies, solutions and methods to attract ownership of the city by adding new energy sources (renewable energy, waste-to-energy solutions, energy efficiency, clean energy, etc. . . .) to the energy portfolio of the city.
- Development of waste-to-energy projects in order to generate energy from wastes, to produce compost from wastes processing, and to improve environmental conditions (land use and pollution).
- Promotion of a wider use of biofuels in transport with the aim of meeting the national strategy and reduce air pollution in Naga City.
- Promotion of energy efficiency and saving activities through expansion of energy audit and conservation campaign to other sectors in order to reduce energy consumption and production, as well as cost of energy.
- Improvement of reliability and quality of electricity supply in order to increase the quality of life of the population, to improve economic performance of industry and services and to increase the interest of entrepreneurs to locate their business in the City.
- Providing full access to electricity to the entire population, including those in remote areas, in order to increase quality of life and attract entrepreneurs to locate their business in these areas.

Motto:

Clean Energy for Quality Life.

Vision:

A vibrant and sustainably developed city with each citizen enjoying quality life supported by clean, affordable and efficiently used energy sources.

Goals:

- Have prepared EIUP and integrated energy plan in the City Medium-term Development Plan in one year.
- Have all municipal waste processed by recycling and by waste-to-energy conversion in case of non-recyclable waste in five years.
- Have at least 5% biofuels in the consumption of motor fuels in the year 2010.
- Have energy audit and conservation campaign expanded to all sectors in five years.
- Have reduced system losses below 10% and reached national standards in power supply quality in 5 years.
- Have 100% connection of households of Naga City in 5 years.

Note: This is an unedited version of the original statement, which is awaiting approval by the City's Executive and Legislative Committees.

Naga City also designed an 'EIUP logo' to increase the visibility and awareness of this new planning activity.

Box 23.5.
EIUP strategy of Can Tho

Eiup Policy:

To solve the problem of lack of energy for future development, the focus will be:

- Implementation of energy planning activities, establishing suitable policies, solutions and methods to attract ownership of the city by adding new energy sources (renewable energy, waste-to-energy solutions, energy efficiency, clean energy, etc. . . .) to the energy structure of city.
- Development of waste-to-energy projects in order to generate energy from wastes, to produce compost from wastes processing, and to improve environmental conditions (land use and pollution).
- Promotion of energy efficiency and saving activities in order to reduce energy consumption and production, as well as cost of energy.
- Development of renewable energy technology to solve the energy supply shortage and meet increasing energy demand caused by the rapid socio-economic development of the city.

Vision:

- Industrial and trade and service activities will have fast grow rate in the future. Moreover, living standards of people will also increase. Therefore, lack of energy, increasing waste, and deteriorating environment will result without suitable planning instruments. These negative impacts will also affect the potential growth rate of the city in the future. Suitable planning will help to limit these negative effects.

Goals:

- Having better coordination between various stakeholders and the city for the development through the establishment of an energy planning unit at city level, able to set-up an energy-environment data base and information system and to prepare city energy planning forecasts; having action plans for the improvement of the city energy systems to tackle future lack of energy and to propose solutions in relation with losses of energy and new sources of energy.
- The city will setup an energy plant for the generation of energy from wastes, production of compost from wastes processing as well as improvement of environmental conditions.
- Preparation and implementation of energy audits and surveys in Industry, commerce, households, agriculture – Identification and assessment of priority sectors and actions for improvement in energy efficiency and savings. Following assessment, preparation and implementation of specific measures for the various sectors of the economy (industry, handicraft, trade and services, households and agriculture).
- Increase additional energy production and increase awareness and participation of private sector in the development of RETs. Moreover, increase capacity of universities and research centers in the field of renewable technologies through RD&D, trainings, etc. . . . and improve environmental standard for production of energy.

Note: This is an unedited version of the original statement, which is currently pending approval by the Can Tho City People's Committee.

Box 23.6.
Action plan for Naga City

Approval of EIUP and inclusion into the City Development Plan

- Conduct stakeholders consultations
- Develop EIUP as a part of the City development plan
- Communicate to the general public and stakeholders to get feedback and support for implementation of activities.

Waste to energy use

- Development of the Feasibility Study for identified priority projects
- Search for potential investors/co-funding
- Priority project implementation.

Promotion of a wider use of biofuels in transport

- Identification of the volume of transport sector activities
- Prepare the plan for marketing of biodiesel and communicate it with the transport sector
- Demonstration of good practice of biodiesel use by cars of the city government
- Promulgation of a city ordinance on the use of biodiesel, public consultation on the ordinance.

Expansion of energy audit and conservation campaign to other sectors

- Identification of potential sectors to be involved in the energy efficiency and auditing campaign
- Issuing of ordinance for implementation of energy efficiency and auditing campaign
- Dissemination of results of activities done so far to demonstrate potential benefits energy efficiency and auditing campaign
- Provide support to other sectors in running energy audits and identification of energy conservation measures
- Presentation of results to show benefits.

Improvement of electricity supply

- Coordination with power supplier in activities to make sure that there is sufficient level of maintenance and monitoring; to monitor and act in minimizing illegal connections or pilferage, and decrease of system losses.

100% electricity supply

- Identification of households not yet connected
- Preparation of connection plan and its implantation
- Identification of financial sources for implementation of the plan.

Note: This is an unedited version of the original statement, which is currently pending approval by the city.

Box 23.7.
Action plan for Can Tho

Development of energy planning and policies at City level

- Creation of an energy planning unit at city level, will need a decision from the People's Committee; however, a realistic budget should be allocated to this unit to perform its activities and duties;
- Creation of an energy and environment Information System and Data Base (energy and environmental statistics and indicators) will require the collaboration and participation of many sectors of the city's economy;
- Prepare yearly energy balances and forecasts, will require that energy companies release information, and that surveys will be conducted by the energy planning unit to know better the energy end-uses in the city;
- Study and propose solutions for future lack of energy (waste-to-energy, renewables, etc. . . .) will also require the cooperation of universities and research centres to study and propose adequate technological solutions;
- Study and propose solutions for energy efficiency and savings measures (regulations, monitoring and enforcement) will require the cooperation of all sectors of the city, including universities and research centres.

Building of a waste treatment plant to produce energy

- An integrated plan for the development of energy in the city should be prepared and proposed to the city's government for action guidelines;
- Establishing pre-feasibility and feasibility studies to be submitted to the city's steering committee. These studies should specify location of plant, chosen technology, financial resources, partners and socio-economic impacts;
- To find partners, investors, international organizations paying attention to electricity production from waste in order to solve the problems (technology, finance, and training);
- Stop or go decision from the municipal administration.

Energy Saving

- Through preliminary integrated survey/audit on energy saving, submit the main results to the people's government and ask for action guidelines;
- Preparing an energy saving programme for the city for the next five years with recommendations up to 2015 to be submitted to the city's government for ratification and implementation;
- Implementing and executing the program with 2 main contents:
- Enhance knowledge and awareness on energy saving for community, government, organizations, businesses, etc. . . .;
- Building demonstration projects on energy saving in residential areas, business, services and industries;
- Establish a steering committee to liaise with communication agency, union, government, etc. . . .

Apply technological achievements to supplement new energy sources being suitable for the city's condition such as solar energy, biogas, biodiesel, etc. . . .

- Writing a report on the necessity and urgency for the city to develop and manufacture new energy technology systems;
- Establishing applied research programmes based on data to be collected, analysed;
- Study existing experiences and benchmarks in the field of renewable energy and organize capacity building;
- Establishing an organization to research and apply technology to the promotion of new energy technologies.

Note: This is an unedited version of the original statement, which is currently pending approval by the Can Tho City People's Committee.

23.7 Conclusions

Prior to participating in the EIUP project, Naga City and Can Tho had built considerable experience in urban environmental management and had some knowledge on energy projects and technologies. Naga City had a solid and international reputation on participative local governance, and both cities had embraced sustainable development principles, even if Can Tho had had limited experience in involving stakeholders. However, they had a vague idea of integrating energy concerns in urban development planning. To start with, reliable data about sectoral energy consumption and future demand were not available. Even national energy policies were seen more as restrictions than a framework under which local initiatives can be drawn. In general the former approaches of the cities in dealing with energy issues were restricted to single projects and a comprehensive framework was missing.

The EU-funded project presented in this chapter has equipped Naga City and Can Tho with the capacity and the tools to tackle energy as an integrated part of their development. This was achieved through the EIUP methodology, which had been used as an instrument for both capacity development and starting the process of integrating energy in urban planning, from building a reliable database to drawing the energy strategies and action plans through a participative process.

Intelligent decision-making builds on reliable and adequate information. So a necessary condition for a successful EIUP is a reliable socioeconomic and energy-environment database. The baseline information generated by this database provide knowledge and understanding of the constraints within which EIUP has to evolve and resources and opportunities through which EIUP goals can be achieved. Another necessary condition for a successful EIUP is knowledge and understanding of the sectors affected by and influencing energy policy. The project has built on the solid experience of Naga City and has built the capacity of Can Tho in identifying and engaging stakeholders.

But what this chapter highlighted as key conditions for a successful EIUP are a strong political commitment on sustainable development and a dedicated and competent team of local experts representing the diverse sectors of the local economy. The first of these conditions cannot be provided by any bilateral or multilateral technical assistance project.

At best, some technical assistance could aim at educating local government administrators and politicians to plant the seed of commitment to sustainable development. Fortunately, a number of local governments (cities and municipalities) have embraced such commitment. And Naga City and Can Tho are glaring examples in this regard. Then it will be easy to achieve the second condition through learning-by-doing and learning from the experiences of other cities. While the actual adherence to the strategies and implementation of the action plans remain to be seen, the two cities can very well share their newly acquired skills and expertise in integrating energy strategies in urban planning.¹¹

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¹¹As indicated in the respective cities' action plans, one important result of this technical assistance project in the two cities is the possibility of developing energy projects from their municipal solid wastes. In fact, the project also taught and discussed with the two cities the possibility of putting up an anaerobic digestion facility to extract biogas and produce electricity at their existing dumpsites. The preliminary assessments have showed the feasibility of such projects, and the two cities are now considering the conduct of full feasibility studies.

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